

Peer Review 1 - Junguk Yoon (*Early Prediction of Chronic Kidney Disease*)

1. Scientific Question

- a. The scientific question was whether machine learning models can predict chronic kidney disease (CKD) risk levels using clinical and drug-related data, especially using non-invasive features. He clearly explained why CKD prediction is important and how clinical variables like blood pressure and diabetes relate to kidney disease. Overall, the motivation and biological relevance were clear.

i. **Rating: Clear (3)**

2. Methodology and Appropriateness

- a. He used PCA and clustering to explore patterns in the data, followed by a Random Forest classifier to predict CKD risk categories. These methods are appropriate because the outcome variable is categorical and tree-based models work well with nonlinear clinical data.

i. **Rating: Appropriate (3)**

- b. They also used bootstrap resampling to estimate confidence intervals, which is a good approach for measuring uncertainty.

i. **Rating: Successfully (2)**

3. Data Visualization

- a. His visualizations were clear and helped communicate the results. The clustering plots and ROC curves with confidence intervals made it easy to understand model performance across different classes. Overall, his figures supported the conclusions of the analysis.

i. **Rating: Great (3)**

4. Interpretation of Results

- a. He concluded that tree-based models performed well for predicting CKD risk and that invasive clinical features improved prediction accuracy. This interpretation was supported by the model results and evaluation metrics shown in the presentation. He also acknowledged limitations such as possible overfitting.

i. **Rating: Well fitted (3)**

Peer Review 2 - Catheryn Inguito, Daniela Arif (*Colorectal Cancer Detection Using DNA Methylation Data*)

1. Scientific Question

- a. The goal of the project was to determine whether DNA methylation patterns can distinguish colorectal cancer samples from normal tissue samples. They explained that methylation changes are common in cancer and may act as biomarkers for disease detection. The biological motivation and importance of the research question were clearly explained.

i. **Rating: Clear (3)**

2. Methodology and Appropriateness

- a. They used PCA to reduce the dimensionality of the methylation dataset and then applied logistic regression and random forest models to classify samples as cancer or normal. These approaches are appropriate because the outcome is binary and PCA helps manage high-dimensional genomic data.

i. **Rating: Appropriate (3)**

- b. They also used bootstrap validation with 1000 resamples, which helps estimate the stability of model performance.

i. **Rating: Successfully (2)**

3. Data Visualization

- a. The UMAP clustering plots helped visualize separation between cancer and normal samples. The logistic regression decision boundary and ROC curve also helped demonstrate classification performance. Overall, the visualizations supported the analysis and were relatively easy to interpret.

i. **Rating: Great (3)**

4. Interpretation of Results

- a. They concluded that methylation patterns differ between colorectal cancer and normal samples and could serve as potential biomarkers. Their models achieved reasonably strong accuracy, which supports this interpretation. They also acknowledged limitations such as biological variability and possible data leakage.

i. **Rating: Well Fitted (3)**